

Software Requirements Specification Weather Forecasting System

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Abstract

The Weather Forecasting System provides users with current weather information and forecasts for a selected location. It provides information such as temperature, humidity, wind speed, and weather conditions.

1 Introduction

1.1 Purpose

The purpose of the system is to provide users with weather information for a selected location. It allows users to check current weather and forecasts.

1.2 Scope

The system allows users to enter or select a location and view its weather information.

The system provides:

- Temperature
- Humidity
- Wind speed
- Weather condition
- Weather forecast

2 Overall Description

2.1 Product Perspective

The Weather Forecasting System receives weather data from a weather data source and presents it to users.

2.2 Product Functions

The main functions are:

1. Select a location.
2. View current weather.
3. View temperature, humidity, and wind speed.
4. View weather forecasts.
5. Update weather information.

2.3 Users

The system is intended for general users who want to check weather conditions and forecasts.

3 Functional Requirements

- The system shall allow users to select a location.
- The system shall display current weather information.
- The system shall provide weather forecasts.
- The system shall update weather information regularly.
- The system shall show an error for an invalid location.

4 Non-Functional Requirements

4.1 Performance

The system should display weather information quickly.

4.2 Usability

The system should have a simple and easy-to-use interface.

4.3 Reliability

The system should provide reliable and updated weather information.

4.4 Security

The system should protect user information from unauthorized access.

5 System Requirements

5.1 Hardware Requirements

- Computer or mobile device
- 2 GB RAM or more
- Internet connection
- Display screen

5.2 Software Requirements

- Windows, Linux, Android, or other supported operating system
- Web browser or supported application
- Internet connection

6 Constraints

- The system requires a reliable weather data source.
- Internet connection may be required for updated information.
- Weather forecasts cannot guarantee exact future conditions.

7 Conclusion

The Weather Forecasting System provides users with current weather information and forecasts. This SRS describes the basic requirements and functionality of the proposed system.